```
#29-12-2025
```

```
#checking datatype
a=12.4
type(a)
```

```
float
```

```
#area of  circle
r=int(input("enter radius:"))
pi=3.14
area=pi*(r*r)
print("area of circle is:",area)
```

```
enter radius:4
area of circle is: 50.24
```

```
#integer operations
a=86
b=9
print("addition:",a+b)
print("subtraction:",a-b)
print("multiplication:",a*b)
print("division:",a/b)
print("floor division:",a//b)
print("modulas:",a%b)
print("exponent:",a**b)
```

```
addition: 95
subtraction: 77
multiplication: 774
division: 9.555555555555555
floor division: 9
modulas: 5
exponent: 257327417311663616
```

```
#string datatype
b="indu"
type(b)
```

```
str
```

```
#boolean datatype
a=7
b=9
print(a>b)
```

```
False
```

```
#list datatype
l=[1,2,3,4]
type(l)
```

```
list
```

```
#tuple datatype
m=[1,2,3,4]
type(m)
```

```
list
```

```
#set datatype
n={1,2,3,4}
type(n)
```

```
set
```

```
#dictt datatype
o={1,2,3,4}
type(o)
```

```
set
```

```
#31-12-2025
```

```
#wap to calculate the SI
p=23567
t=4
r=2
si=(p*t*r)/100
print("simple intrest is :",si)
```

```
simple intrest is : 1885.36
```

```
#wap to calculate the average of 4 sub
m1=52
m2=56
m3=89
m4=52
avg= (m1+m2+m3+m4)/4
print("avg of 4 sub is:",avg)
```

```
avg of 4 sub is: 62.25
```

```
#wap to calculate power of number
a=9
b=3
print("power of number is:",a**b)
```

```
power of number is: 729
```

```
#wap to print current bill
unit=7
price=10
bill=unit*price
print("current bill is:",bill)
```

```
current bill is: 70
```

```
#wap to check wheather num is positive or negative in boolean values without using if
a=8
print(a>0)
```

```
True
```

```
#wap to print given num lies between 10 to 20
n=18
print("given" ,n ,"lies in btwn 10to 20",n>10 and n<=20)
```

```
given 18 lies in btwn 10to 20 True
```

```
#take a number >10 check if unit digit of that number is even or not
n=168
a=n%10
print(a%2==0)
```

```
True
```

```
#2-01-2026
```

```
name='indu'
print(name)
```

```
indu
```

```
#complex number
a=1j+2j
a
```

```
3j
```

```
#comparison
a=10
b=15
print(a<b)
```

```
True
```

```
#list
List=[1,2,3,4,5,6,6,6,6,66]
print(List)
```

```
[1, 2, 3, 4, 5, 6, 6, 6, 6, 66]
```

```
#tuple
colors=(5,6,7,8,9,)
print(colors)
```

```
(5, 6, 7, 8, 9)
```

```
#set
Colors={4,5,6,7,8,8,9,9,8,8,9}

print(Colors)
```

```
{4, 5, 6, 7, 8, 9}
```

```
#3-01-2026
```

```
#print name
First="KANUGALA "
Middle="indu"
Last="mathi"
print(First+Middle+Last)
```

```
KANUGALA indumathi
```

```
#arithmetic operator
a=10
b=6
c=a*b
print("multiplication:",c)
print("division:",a/b)
print("division:",a//b)
```

```
multiplication: 60
division: 1.6666666666666667
division: 1
```

```
#set
Set={"green","black","orange"}
print (Set)
```

```
{'orange', 'green', 'black'}
```

```
#dict
Student={"name":"indu","age":21,"group":"BCA"}
print(Student)
```

```
{'name': 'indu', 'age': 21, 'group': 'BCA'}
```

```
#to print area of circle
r=4
area=3.14*r*r
print("area of circle:",area)
```

```
area of circle: 50.24
```

```
#logical operator
```

```
25<10 and 35>10
```

```
False
```

```
25<10 and 35<10
```

```
False
```

```
25>10 and 35>10
```

```
True
```

```
not 25<10
```

```
True
```

```
25<10 or 35<10
```

```
False
```

```
25>10 or 35>10
```

```
True
```

```
25>10 or 35<10
```

```
True
```

```
#wap to calculate SI
P=1000000
T=0.5
R=2
Si=(P*T*R)/100
print("simple intrest is :",Si)
```

```
simple intrest is : 10000.0
```

```
#wap to calculate avg marks
M1=56
M2=65
M3=85
M4=98
Avg=(M1+M2+M3+M4)/4
print("avg of 4 sub is :",Avg)
```

```
avg of 4 sub is : 76.0
```

```
#exponent/ power
A=2
B=4
print(A**B)
```

```
16
```

```
#wap to calculate current bill
Units=10
Price=6
Bill=Units*Price
print("current bill:",Bill)
```

```
current bill: 60
```

```
#print +ve number without using loops / if
A=5
print(A>0)
```

```
True
```

```
#print num using operators
n=19
print("given number",n,"lies btwn 10 and 20:",n>10 and n<20)
```

```
given number 19 lies btwn 10 and 20: True
```

```
#even or odd using  operators
n=5
n>0
a=n%10
print(a%2==0)
```

```
False
```

```
#using input
```

```
#10-01-2026
```

```
a=int(input(" enter a :"))
b=int(input(" enter b :"))
print("sum:",a+b)
print("difference:",a-b)
print("mult:",a*b)
print("div:",a/b)
print("floor:",a//b)
print("modular :",a%b)
print("exp:",a**b)
```

```
 enter a :5
 enter b :4
sum: 9
difference: 1
mult: 20
div: 1.25
floor: 1
modular : 1
exp: 625
```

```
a=int(input("enter a :"))
b=int(input("enter b:"))
print(a>b and b<a)
print(a>b or b<a)
print(a<b and b<a)
print(a<b and b>a)
print(a>b or b>a)
print(a<b and b<a)
print(not a>b )
```

```
enter a :56
enter b:75
False
False
False
True
True
False
True
```

```
m1=int(input("enter m1 marks :"))
m2=int(input("enter m2 marks :"))
m3=int(input("enter m3 marks :"))
print(m1>=35 and m2>=35 and m3>=35)
```

```
enter m1 marks :45
enter m2 marks :67
enter m3 marks :25
False
```

```
m1=int(input("enter m1 marks :"))
m2=int(input("enter m2 marks :"))
m3=int(input("enter m3 marks :"))
print(m1>=35 == m2>=35 != m3>=35)
```

```
enter m1 marks :1
enter m2 marks :56
enter m3 marks :57
False
```

```
#to print pass or fail
m1=int(input("enter marks:"))
m2=int(input("enter marks:"))
m3=int(input("enter marks:"))
print((m1>=35)and(m2>=35) or(m3>=35))
```

```
enter marks:30
enter marks:45
enter marks:55
True
```

```
#biggest among three
a=int(input("enter a:"))
b=int(input("enter b:"))
c=int(input("enter c:"))
d=int(input("enter d:"))
if(a>b and a>c and a>d):
     print(" a is big")
elif(b>a and b>c and b>d):
     print(" b ig big ")
elif(c>a and c>b and c>d):
     print(" c ig big ")
else:
     print(" d is big")
```

```
enter a:56
enter b:67
enter c:78
enter d:76
 c ig big
```

```
#branching statements
```

```
#11-01-2026
```

```
m=int(input(" enter marks:"))
if m>=0 and m<=100:
    if m>=90:
        print(" grade A")
    elif m>=75:
        print(" grade B")
    elif m>=60:
        print(" grade C")
    elif m>=40:
        print(" grade D")
    else:
        print(" Fail")
else:
    print(" invalid marks")
```

```
 enter marks:89
 grade B
```

```
m=int(input(" enter marks:"))
if 0<=m<=100:
    if m>=90:
        print(" grade A")
    elif m>=75:
        print(" grade B")
    elif m>=60:
        print(" grade C")
    elif m>=40:
        print(" grade D")
```

```
    else:
        print(" Fail")
else:
    print(" invalid marks")
```

```
 enter marks:678
 invalid marks
```

```
a=int(input(" enter side a:"))
b=int(input(" enter side b:"))
c=int(input(" enter side c:"))
if a+b >c and b+c>a and a+c>b:
   if a==b and a==c and b==c:
        print(" equilateral triangle")
   elif a==b or b==c or c==a:
        print(" isosceles triangle")
   else:
        print(" scalen")
else :
    print("invalid triangle")
```

```
 enter side a:78
 enter side b:67
 enter side c:56
 scalen
```

```
salary=int(input(" enter salary"))
yoe=int(input(" enter your experience:"))
if salary<20000 and yoe>=2:
    b=salary*0.1
    amount=salary+b
    print( amount)
elif salary>=20000 and yoe >=5:
    c=salary*0.2
    amount=salary+c
    print(amount)
else:
    print(" no bonus")
```

```
 enter salary22000
 enter your experience:5
26400.0
```

```
n=int(input(" enter a num :"))
if n%3==0 and n%5!=0:
    print(" spcl num...")
else:
    print(" not a spcl num...")
```

```
 enter a num :11
 not a spcl num...
```

```
t=int(input("enter time:"))
if t>=0 and t<=24:
    if t>=5 and t<=11 :
        print(" good morning…")
    elif t>=12 and t<=16:
        print(" good Afternoon…")
    elif t>=17 and t<=20:
        print(" good evening")
    else:
        print(" good night")
else:
    print(" invalid")
```

```
enter time:0
 good night
```

```
a=int(input(" enter first num:"))
b=int(input("enter second num:"))
c=int(input("enter third num:"))
if a>b and a<c or a<b and a>c:
    print(a ,"is middle ")
elif b>a and b<c or b<a and b>c :
    print(b,"is middle")
```

```
else:
    print(c ,"is middle ")
```

```
 enter first num:12
enter second num:14
enter third num:13
13 is middle
```

```
a=int(input("enter a:"))
b=int(input("enter b:"))
c=int(input("enter c:"))
d=int(input("enter d:"))
if(a>b and a>c and a>d):
     print(" a is big")
elif(b>a and b>c and b>d):
     print(" b ig big ")
elif(c>a and c>b and c>d):
     print(" c ig big ")
else:
     print(" d is big")
```

```
enter a:45
enter b:67
enter c:78
enter d:42
 c ig big
```

```
#22-1-26
```

```
#while lioop
```

```
n=int(input(" enter n:"))
i=1
while i<=n:
    print(i)
    i=i+1
```

```
 enter b:10
1
2
3
4
5
6
7
8
9
10
```

```
n=int(input(" enter n:"))
i=0
while i<=n:
    print(i)
    i=i+2
```

```
 enter n:10
0
2
4
6
8
10
```

```
n=int(input(" enter n:"))
i=1
sum=0
while i<=n:
    sum=sum+i
    i=i+1
print(sum)
```

```
 enter n:10
55
```

```
#23-1-26
```

```
#for loop
```

```
for i in range (0,20,2):
    print(i)
```

```
0
2
4
6
8
10
12
14
16
18
```

```
for i in range (1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

```
for i in range (3,31,3):
    print(i)
```

```
3
6
9
12
15
18
21
24
27
30
```

```
for i in range (1,11):
    print(i*3)
```

```
3
6
9
12
15
18
21
24
27
30
```

```
for i in range(10,0,-1):
    print(i)
```

```
10
9
8
7
6
5
4
3
2
1
```

```
#interview oriented
```

```
name="indu"
for s in name:
    print(s)
```

```
i
n
d
u
```

```
n=356
q=n%10
print(q)
```

```
6
```

```
n=356
q=n//10
print(q)
```

```
35
```

```
n=3456
count=0
while n>0:
    n=n//10
    count =count+1
print(count)
```

```
4
```

```
#rev a num
n=3456
rev=0
while n>0:
    rem=n%10
    rev=(rev*10)+rem
    n=n//10
print(rev)
```

```
6543
```

```
#sum of a num
n=456
sum=0
while n>0:
   m=n%10
   sum=sum+m
   n=n//10
print(sum)
```

```
15
```

```
#rev a num
n=int(input("enter a num:"))
temp=n
rev=0
while n>0:
    rem=n%10
    rev=(rev*10)+rem
    n=n//10
if temp==rev:
    print(" palindrome")
else:
    print(" not a palindrome")
```

```
enter a num:5665
 palindrome
```

```
for i in range (5):
    if i ==5:
         Break
    print(i)
```

```
0
1
2
3
4
```

```
#24-1-26
```

```
#functions
```

```
def add(a,b):
   return a+b


add(7,6)
```

```
13
```

```
def myfun(x,y=50):
    print("x:",x)
    print("y:",y)
myfun(20)


myfun(10,60)
```

```
x: 20
y: 50
x: 10
y: 60
```

```
def greet():
    print(" welcome to python programming...")

greet()
```

```
 welcome to python programming...
```

```
def add(a,b):
    print(a+b)


add(34,7)
```

```
41
```

```
def even(x):
    if x%2==0:
       print("even ")
    else:
       print("odd")

even(8)
```

```
even
```

```
def square(a):
    print(a**2)

square(27)
```

```
729
```

```
def details(name,age,group):
    print("name:",name)
    print("age:",age)
    print("group:",group)

details("indu",21,"BCA")
```

```
name: indu
age: 21
group: BCA
```

```
def greet():
    return(" welcome to python programming...")

print(greet())
```

```
 welcome to python programming...
```

```
def add(a,b):
    return(a+b)


print(add(34,7))
```

```
41
```

```
def even(x):
    if x%2==0:
       return ("even ")
    else:
       return("odd")
print(even(8))
```

```
even
```

```
def square(a):
    return(a**2)

print(square(27))
```

```
729
```

```
def fact(n):
    if n==0:
        return 1
    else:
        return n * fact(n-1)

fact(4)
```

```
24
```

```
def details(name,age,group):
    return "name:",name,"age:",age,"group:",group

print(details("indu",21,"BCA"))
```

```
('name:', 'indu', 'age:', 21, 'group:', 'BCA')
```

```
#scope of variables
```

```
#local
def show():
    x=10
    print(x)
show()
```

```
10
```

```
#global
x=15
def show():
   print(x)
show()
```

```
15
```

```
def show():
    c=10
show()
def display():
    print(c)
display()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipython-input-1524872493.py in <cell line: 0>()
      4 def display():
      5     print(c)
----> 6 display()

/tmp/ipython-input-1524872493.py in display()
      3 show()
      4 def display():
----> 5     print(c)
      6 display()

NameError: name 'c' is not defined
```

```
#29-1-26
```

```
def display(name, age):
    return f"Name:{name}\nAge:{age}"

print(display("Suraj", 21))
```

```
Name:Suraj
Age:21
```

```
name='Indu'
for char in name:
    print(char,end=" ")
```

```
I n d u
```

```
s=" welcome to python "
print(s[::-1])
```

```
 nohtyp ot emoclew
```

```
s=" welcome to python "
print(s[1:5])
```

```
welc
```

```
s=" welcome to python "
print(s[:8])
```

```
 welcome
```

```
s="welcome"
print(s[::-2])
```

```
eolw
```

```
len(s)
```

```
7
```

```
print(max(s))
```

```
w
```

```
print(min(s))
```

```
c
```

```
K=" Welcome to python "
print(K.upper())
```

```
 WELCOME TO PYTHON
```

```
K=" Welcome to python "
print(K.lower())
```

```
 welcome to python
```

```
K=" Welcome to python "
print(K.title())
```

```
 Welcome To Python
```

```
K=" welcome to python "
print(K.capitalize())
```

```
 welcome to python
```

```
#array
```

```
from array import array
array=array('i',[1,2,3,4])
print(type(array))
```

```
<class 'array.array'>
```

```
print(array)
```

```
array('i', [1, 2, 3, 4])
```

```
print(array[2:])
```

```
array('i', [3, 4])
```

```
print(array[::-1])
```

```
array('i', [4, 3, 2, 1])
```

```
print(array[::-2])
```

```
array('i', [4, 2])
```

```
print(array[::2])
```

```
array('i', [1, 3])
```

```
array[1]=25
print(array)
```

```
array('i', [1, 25, 3, 4])
```

```
import array as arr
arr=arr.array('i',[1,2,3,45])
print(arr)
type(arr)
```

```
array('i', [1, 2, 3, 45])
array.array
```

```
for x in arr:
    print(x)
```

```
1
2
90
45
50
```

```
#array methods
```

```
arr.append(50)
print(arr)
```

```
array('i', [1, 2, 3, 45, 50])
```

```
arr.insert(3,90)
print(arr)
```

```
array('i', [1, 2, 3, 90, 45, 50])
```

```
arr.removr(3)
print(arr)
```

```
array('i', [1, 2, 90, 45, 50])
```

```
arr.pop()
print(arr)
```

```
array('i', [1, 2, 90, 45])
```

```
print(arr.index(45))
```

```
3
```

```
print(arr.count(2))
```

```
1
```

```
#1
import array as a
a=a.array('i',[1,2,3,45])
for i in a:
    print(i)
```

```
1
2
3
45
```

```
#2
count=0
import array as a
a=a.array('i',[1,2,3,45])
for i in a:
    count+=1
print(count)
```

```
4
```

```
#3
sum=0
import array as a
a=a.array('i',[1,2,3,45])
for i in a:
    sum+=i
print(sum)
```

```
51
```

```
#4
sum=0
import array as a
a=a.array('i',[1,2,3,45])
c=a[::2]
print(c)
for i in c:
    sum+=i
print(sum)
```

```
array('i', [1, 3])
4
```

```
#5
list=[1,2,3,45]
large=list[0]
for i in list :
    if i>large:
```

```
        large=i
print(large)
```

```
45
```

```
#6
list=[1,2,3,45]
large=list[0]
for i in list :
    if i<large:
        large=i
print(large)
```

```
1
```

```
#7
count=0
import array as a
a=a.array('i',[4,2,3,44])
for i in a:
    if i%2==0:
        count+=1
    else:
        count+=1
print("even no count:",count)
print("odd no count:",count)
```

```
even no count: 4
odd no count: 4
```

```
count=0
import array as a
a=a.array('i',[4,2,3,44])
for i in a:
    if i%2==0:
      print("odd no count:",c)
    else:
      print("even no count:",count)
```

```
odd no count: 1
odd no count: 1
even no count: 0
odd no count: 1
```

Start coding or generate with AI.